MILITARY SPECIFICATION

INSERT MOUNTING UNIT, MEDICAL INSTRUMENT AND SUPPLY SET CHEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an insert mounting unit, suitable for insertion in the Standard No. 6 medical service aluminum field chest, and a tabletop unit suitable for attachment to No. 6 field chests.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-225/6	Aluminum Alloy, Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024
QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/8	Aluminum Alloy 5052, Plate and Sheet
QQ-M-31	Magnesium Alloy, Bars, Rods and Special Shaped Sections, Extruded
QQ-S-766	Steel Plate, Sheet, and Strip-Corrosion Resisting
NN-P-530	Plywood, Flat Panel
TT-E-527	Enamel, Alkyd, Lustreless.

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TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal
TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
TT-T-306	Thinner, Synthetic Resin, Enamels
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion On
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-W-13518	Wood Preservative, Tetrachlorophenol And Pentachlorophenol, Surface Sealing Compound
MIL-P-15328	Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals)
MIL-C-16775	Chest, Field, Aluminum, Empty

STANDARDS

Federal

FED-STD-151	Metal, Test Methods
FED-STD-595	Color

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Military

MIL-STD-129	Marking for Shipment and Storage
MS16686	Medical Services Symbol

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DRAWING

Defense Personnel Support Center (DPSC)

10018 Inserts for Dental Equipment Set, Prosthetic, Field.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawing (with Bill of Materials) referenced herein, and shall be capable of meeting the requirements shown therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in Section 4. When materials to be used are not specifically designated, those uded shall be of the highest commercial quality, and suitable for the purpose intended.

3.1.1 Aluminum sheet. Aluminum alloy sheet shall conform to QQ-A-250/8, and the temper shall be as indicated on the Bill of Material. Part Nos. 31, 32, 33, 35, 37, 38, 39, 42, 43 and 51 shall be fabricated of this material.

3.1.2 Aluminum bar. Aluminum alloy bar shall conform to QQ-A-225/6, and the temper shall be as indicated on the Bill of Material. Part Nos. 40 and 47 shall be fabricated of this material. As an alternate, Part Nos. 40 and 47 shall be fabricated of sheet conforming to QQ-A-250/4; temper shall be as indicated on Bill of Material.

3.1.3 Steel, corrosion-resisting. Corrosion resisting steel shall conform to QQ-S-766, class 302 or 304. Part Nos. 34, 36, 41, 44 and 46 shall be fabricated of this material.

3.1.4 Magnesium, extruded. Extruded magnesium alloys shall conform to QQ-M-31. Part Nos. 53 through 58 shall be fabricated of this material.

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3.1.5 Plywood. Plywood shall conform to NN-P-530, Group B, Type I, Exterior.

3.1.6 Wood Preservative. Wood preservative shall conform to MIL-W-13518, Type I, class 1.

3.1.7 Primer Coating. Primer coating for the plywood shall conform to TT-P-636.

3.1.8 Primer. Primer for metals shall conform to TT-P-1757, and shall be Color Y.

3.1.9 Enamel. Enamel shall conform to TT-E-527. Color of the enamel shall approximate 34087 (Lusterless olive drab) as shown in FED-STD-595, except that for marking as specified in 3.5, the following colors of the same standard shall be used (see 6.3):

Black	37038
Red	31136
White	37875
Blue	35044

3.2 Component parts.

3.2.1 Metal mounting unit. The metal mounting unit consists of a false bottom, a casting machine mounting, a grinding and polishing machine mounting, and a cradle mounting for holding two cylinders of acetylene.

The false bottom, part No. 31, is secured into the chest by means of four bracket screws, part No. 45, passing through the corner brackets, part No. 44, of the false bottom. The corner brackets are riveted to the false bottom and having backing strips, part No. 46.

The mounting for holding two 10-cubic foot cylinders of acetylene is welded to the false bottom and consists of a cradle, part No. 32, with two shelves, part No. 33; a "U" bolt, part No. 34, and two "U" bolt clamps, part No. 37; and a cylinder stop, part No. 35.

The mounting for securing a casting machine to the false bottom consists of three clips, part No. 36; a retainer, part No. 38; and a holder, part No. 39.

The mounting for holding the grinding and polishing machine is welded to the false bottom and consists of two "V" bolts, part No. 41; two "V" bolt bars, part No. 42; and two "V" bolt clamps, part No. 43.

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3.1.2 Plywood tabletop. The tabletop subassembly consists of four plywood sections (one each of part Nos. 48 and 50; and two sections of part No. 49); two clamps part No. 47; two chest lamps, part No. 51, 6 latches, part No. 40; and latch retainers, part Nos. 53 through 58 inclusive. The tabletop shall provide a flat working area of approximately 27 inches wide by 60 inches long.

3.3 Construction. Construction shall conform in all details to the requirements specified herein and as shown on applicable portions of the Drawing (with Bill of Materials). The construction shall be sound and durable in all respects. The assembly shall be welded on jigs capable of maintaining required tolerances on finished mounting units. The insert mounting unit shall fit into a No. 3 medical field chest conforming to MIL-C-16775 (see 6.4), when tested as specified in 4.3.2. The tabletop shall not sag more than 2 inches at the midpoint from the horizontal plane of the span when tested as specified in 4.3.3.

3.3.1 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter; any cleaning method will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. These welds shall be continuous with no visible voids, pin holes, or cracks. Welds shall not be oxidized.

3.4 Finish.

3.4.1 Aluminum alloy. After welding has been completed, but before final assembly, all accessible aluminum surfaces of the insert mounting unit shall be finished with enamel as specified herein. Prior to all application of the finish, all surfaces shall be thoroughly cleaned, free from dirt, oil, grease, welding flux and loose scale with solvents, detergents, or other processes which have no deleterious effect on the surfaces. After thorough cleaning and degreasing, surfaces to be enameled shall be prepared for topcoating by one of the following methods:

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(1) Chemical treatment in accordance with MIL-C-5541.
(2) Anodic treatment in accordance with MIL-A-8625.
(3) Priming treatment, using primer specified in 3.1.8, thinned and applied by spray in accordance with the directions specified under "Preparation for Delivery" of TT-P-1757, and air-dried for not less than 30 minutes before top-coating.
(4) Preprimining with a wash-primer coat in accordance with MIL-P-15328, and then treated as in (3) above.

Topcoating shall be with one coat of enamel as specified in 3.1.9. The enamel shall be reduced by taking four parts by volume of enamel with one part by volume of thinner in conformance with TT-T-306. The reduced enamel shall be applied by spraying and cured by a heat treatment equivalent to baking at 250 Degree Fahrenheit for not less than 45 minutes. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, cracking, acid, alkali, and solvents.

3.4.2 Magnesium alloy. Magnesium parts shall be corrosion preventive treated in accordance with type III (except alloy MIA) or type IV of MIL-M-3171. Magnesium parts shall be then given one coat of primer as specified in 3.1.8, thinned and applied by spray in accordance with the directions specified under "Preparation for Delivery" of TT-P-1757, and dried for not less than 30 minutes, and shall then be given one coat of enamel as specified in 3.1.9.

3.4.3 Plywood. Before attachment of metal thereto, all surfaces of plywood shall be sanded smooth, given one coat of wood preservative, see 3.1.6; then one coat of primer coating, see 3.1.7; and finally one coat of enamel, see 3.1.9.

3.4.4 Steel. After riveting to the false bottom, the exposed surfaces of the false bottom bracket, part No. 44, and false bottom backing strip, part No. 46, shall be cleaned and degreased by a process which shall have no deleterious effect on the surfaces, and shall be primed as specified in method 3 of 3.4.1, and then finished with enamel as specified in 3.1.9.

3.4.5 Film thickness of paint. Contractor shall endeavor to control film thickness so that the thickness of a single dried coat of paint is 0.3 to 0.7 mil for primer and 1.0 to 1.5 mil for enamel.

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3.5 Identification markings.

3.5.1 Insert mounting unit. The insert mounting unit shall be permanently and legibly marked with black enamel, as specified in 3.1.9, on the underside of the false bottom, as follows:

a. A one inch medical services symbol conforming to type I of MS 16686.

b. The following information shall be included:

6545-00-531-7838	3/8" High
INSERT MOUNTING UNIT, MEDICAL INSTRUMENT AND SUPPLY SET CHEST	3/8" High
Name or registered trademark of the manufacturer	1/4" High
Contract Number	1/4" High

3.5.2 Plywood tabletop. On the underside of the plywood tabletop sections, rectangular strips to facilitate assembly, shall be marked with enamel as shown on the drawing, in the appropriate colors as shown, as specified in 3.1.9. In addition, in vertical block letters 1/4 inch high, there shall be marked with black enamel, (see 3.2.9), the legend: "To assemble tabletop sections, match rectangular color strips of the same color. Four sections make up one tabletop."

3.6 Workmanship. Workmanship shall be first class throughout. Insert mounting unit shall be square and free from twist. All burrs shall be removed and all sharp corners shall be broken. Completed insert mounting units and plywood tabletops shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three (3) years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the aluminum sheet and bar, magnesium shapes, corrosion-resisting steel, and plywood, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of these items or lots of items to be offered for acceptance are classified as specified in tables I and II (see 4.2.1, 4.2.2) and 4.2.3.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-129. The inspection level and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one insert mounting unit.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor A	II	4.0
Minor B	S-1	1.0

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4.2.2 Examination.

4.2.2.1 Mounting unit. The insert mounting unit shall be examined for defects including, but not limited to, those listed in table II:

Table II. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Unit not complete with all specified component parts.
102	Any component parts not free of a crack, tear, rupture, and deformed surface.
103	Any exposed or accessible edge not free of a burr and sharpness.
104	Any weld not sound and smoothly finished.
105	Finish not smooth, even, and adherent.
106	Unit identification markings not complete and correct.
Minor A	
201	Identification marking not completely legible.
202	Enameled surfaces not free of chipped and scratched areas.
203	Color of enamel not as required.
204	Not clean, free of grease, oil, dirt, other foreign matter.
Minor B	
251	Dimensions not within tolerances shown on drawing.

4.2.2.2 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 (defects per hundred units).

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Table III. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, Inspection Level S-1, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one insert mounting unit.

4.2.3.1 Sampling for chemical testing of metals. A sufficient amount of each lot of each metal used in the construction of the cases shall be selected for chemical analysis as specified in 4.3.1.

4.3 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.3.1 Chemical analysis of metals. The samples selected in accordance with 4.2.3.1 shall be analyzed in conformance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

4.3.2 Fit test. The metal mounting unit shall be fully assembled, and then positioned in a No. 3 medical field chest (see 6.4), and secured in place with the four bracket screws.

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4.3.3 Deflection test. The four plywood sections of the tabletop subassembly shall be assembled in accordance with instructions. The complete tabletop shall be supported along the approximate longitudinal centerline of the outside sections. Minimum bottom clearance shall be 6 inches. A ten pound concentrated weight shall be placed on approximately the geometric centerlines of the tabletop, and the deflection measured.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each insert mounting unit shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.1.2 Level C. Each insert mounting unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Six (6) insert mounting units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3, PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Six (6) insert mounting units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1 or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Insert mounting units shall be in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The completed insert mounting unit is intended to hold part of the contents of NSN 6545-00-299-8593, DENTAL EQUIPMENT SET, PROSTHETIC, FIELD, Part B. The tabletop is intended for use with both NSNs 6545-00-299-8592, DENTAL EQUIPMENT SET, PROSTHETIC, FIELD, Part A, and 6545-00-299-8593, DENTAL EQUIPMENT SET, PROSTHETIC, FIELD, Part B, as shown on sheet 1 of the drawing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number and Item Identification.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Availability of color chips. Individual chips of the colors specified in 3.1.9 are available from:

Commissioner
Federal Supply Service
General Services Administration
Standardization Division
Paints Branch
Washington, DC 20406

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6.4 Sample chest. A 6545-00-914-3480, No. 3 Medical Instrument and Supply Set Field Chest may be obtained on a loan basis upon application to the Contracting Officer and payment of the required deposit.

6.5 The insert mounting unit covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock Number	Item Identification
6545-00-531-7838	INSERT MOUNTING UNIT, MEDICAL INSTRUMENT AND SUPPLY SET CHEST

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project number: 6545-0026

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-37239(DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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Philadelphia, PA. 19101

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DEFENSE SUPPLY AGENCY
DoD - 304



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PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

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